

WATROUS EAST, SK, Canada

WMO: 715110

Lat: 51.670N

Lon: 105.400W

Elev: 1724

StdP: 13.80

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-21.9	-33.0	0.9	-27.0	-27.4	1.3	-21.9	29.3	14.9	26.4	15.8	7.7	300	0.731

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	85.5	65.3	81.9	64.3	78.8	62.7	70.1	79.8	67.7	76.7	65.6	74.5	11.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	106.0	75.0	64.5	97.0	72.2	62.2	89.3	69.7	35.3	79.7	33.2	76.7	31.4	74.6	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
25.6	22.8	20.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-35.0	92.8	6.0	3.3	-39.3	95.2	-42.8	97.1	-46.2	98.9	-50.5	101.3
			-35.1	74.8	5.9	2.3	-39.4	76.4	-42.8	77.7	-46.1	79.0	-50.3	80.6
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	4.7	7.6	19.8	37.7	50.6	59.7	64.2	62.4	53.5	38.9	22.2	9.1	(6)
(7)		DBStd	24.08	15.72	14.39	14.13	10.21	8.36	6.23	5.17	6.00	7.85	8.87	12.17	14.22	(7)
(8)		HDD50	6518	1403	1188	936	380	96	4	0	1	52	355	835	1268	(8)
(9)		HDD65	10700	1868	1608	1401	820	449	178	77	122	351	808	1285	1733	(9)
(10)		CDD50	1415	0	0	0	11	115	296	440	385	156	12	0	0	(10)
(11)		CDD65	122	0	0	0	0	3	21	52	41	5	0	0	0	(11)
(12)		CDH74	2033	0	0	0	6	133	316	660	687	226	5	0	0	(12)
(13)		CDH80	522	0	0	0	2	27	71	156	206	60	0	0	0	(13)
(14)	Wind	WSAvg	9.9	10.2	10.0	10.4	10.9	11.4	10.1	8.5	8.4	9.6	10.0	10.0	9.6	(14)
(15)	Precipitation	PrecAvg	16.3	0.7	0.4	0.6	1.1	1.9	3.3	2.6	2.1	1.5	1.0	0.6	0.6	(15)
(16)		PrecMax	28.5	1.4	0.8	1.5	2.4	5.2	6.8	5.7	7.4	5.0	2.9	1.9	1.4	(16)
(17)		PrecMin	8.9	0.1	0.1	0.1	0.0	0.4	0.6	0.7	0.4	0.1	0.0	0.1	0.2	(17)
(18)		PrecStd	4.1	0.4	0.2	0.4	0.7	1.0	1.5	1.3	1.5	1.1	0.7	0.4	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	39.2	53.0	71.4	83.7	87.0	88.9	91.4	86.3	72.7	53.7	41.3	(19)
(20)		0.4%	MCWB	36.2	35.6	44.0	52.4	58.4	63.6	70.6	66.1	62.3	56.6	45.1	36.3	(20)
(21)		2%	DB	36.5	35.0	45.2	66.1	77.7	81.4	84.1	85.7	80.7	64.4	45.8	37.0	(21)
(22)		2%	MCWB	33.9	32.8	39.8	49.9	55.8	62.5	68.9	64.8	59.9	51.3	39.6	34.3	(22)
(23)		5%	DB	32.7	31.9	40.4	60.6	73.0	77.3	80.7	81.3	75.1	58.1	40.9	33.0	(23)
(24)		5%	MCWB	30.7	30.6	36.9	47.8	53.9	60.8	66.7	64.0	58.0	47.9	37.1	31.0	(24)
(25)		10%	DB	27.2	28.3	36.7	54.7	68.4	73.8	77.5	77.4	69.7	52.9	37.4	28.6	(25)
(26)		10%	MCWB	26.1	27.1	34.6	45.0	52.0	58.9	65.4	62.3	55.5	45.6	34.9	27.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	36.1	44.5	54.8	61.7	70.6	75.3	71.4	65.6	57.0	45.7	36.9	(27)
(28)		0.4%	MCDWB	39.1	38.4	51.1	67.8	73.1	80.6	83.6	83.4	79.4	70.6	53.3	40.3	(28)
(29)		2%	WB	34.0	33.0	40.2	51.7	58.3	66.5	71.6	68.7	62.2	52.4	40.4	34.2	(29)
(30)		2%	MCDWB	36.3	34.7	44.7	63.5	71.3	74.7	80.9	79.4	75.4	60.6	44.5	36.7	(30)
(31)		5%	WB	30.6	30.5	37.1	48.7	56.0	63.5	68.9	66.5	59.4	49.1	37.5	31.0	(31)
(32)		5%	MCDWB	32.6	31.9	40.1	58.3	69.7	72.3	77.4	76.8	71.6	56.4	40.4	33.0	(32)
(33)		10%	WB	26.1	27.1	34.7	45.5	53.7	61.2	66.7	64.3	56.8	46.3	34.9	27.2	(33)
(34)		10%	MCDWB	27.1	28.3	36.7	53.7	65.4	70.1	74.9	73.6	67.8	52.7	37.5	28.6	(34)
(35)	Mean Daily Temperature Range		MDBR	17.6	17.9	17.5	21.1	24.8	21.2	22.4	24.7	24.6	20.2	16.6	16.6	(35)
(36)		5% DB	MCDDBR	19.9	18.7	18.9	32.4	32.5	28.5	28.6	32.9	35.6	30.8	21.7	19.8	(36)
(37)			MCWBR	17.9	16.8	14.2	17.9	14.8	13.1	15.3	14.8	17.0	18.3	16.1	17.3	(37)
(38)			MCDDBR	19.6	18.3	18.4	29.8	28.1	23.7	25.5	27.0	31.2	27.9	20.7	19.9	(38)
(39)		5% WB	MCWBR	17.7	16.5	14.2	17.8	14.0	12.6	15.3	14.3	15.9	17.3	15.7	17.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.263	0.293	0.328	0.342	0.355	0.361	0.365	0.316	0.284	0.251	0.233	(40)	
(41)		taud	2.304	2.276	2.349	2.388	2.397	2.408	2.390	2.373	2.495	2.545	2.480	2.358	(41)	
(42)		Ebn at Noon	247	276	289	289	289	285	281	273	278	269	251	239	(42)	
(43)		Edh at Noon	20	27	31	34	35	35	35	34	27	21	16	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	377	712	1183	1530	1870	1905	1993	1613	1185	689	346	266	(44)	
(45)		RadStd	36	47	102	128	157	135	114	115	97	96	48	27	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page